

Sequences & Series

ANSWER KEY



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10} .

$$a_{10} = a_1 + (10 - 1)d = 9 + 9(6) = 63.$$

- 2 For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11} .

$$a_{11} = 15 + 10(-4) = 15 - 40 = -25.$$

Arithmetic series

- 3 Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).

$$S_{10} = \frac{n(a_1 + a_n)}{2} = \frac{10(9 + 63)}{2} = 5(72) = 360.$$

Geometric sequences

- 4 For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8 .

$$a_8 = a_1 r^7 = 6(2^7) = 6(128) = 768.$$

- 5 A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.

$$a_n = 2 \cdot 5^{n-1}. \text{ Setting } 2 \cdot 5^{n-1} = 1250 \text{ gives } 5^{n-1} = 625 = 5^4, \text{ so } n - 1 = 4 \text{ and } n = 5.$$

Geometric series (finite)

- 6 Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).

$$S_8 = \frac{a_1(r^8 - 1)}{r - 1} = \frac{6(256 - 1)}{2 - 1} = 6(255) = 1530.$$

Infinite geometric series

- 7 Find the sum of the infinite geometric series $36 + 12 + 4 + \dots$.

$$a = 36, r = \frac{1}{3}, \text{ and } |r| < 1, \text{ so the sum is } \frac{36}{1 - \frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$$

- 8 Find the sum of the infinite geometric series $12 - 4 + \frac{4}{3} - \dots$.
- $a = 12$, $r = -\frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{12}{1 - (-\frac{1}{3})} = \frac{12}{\frac{4}{3}} = 9$.